

# Software Project Management

## EC-431

### Topics and Subtopics of

#### *1. Introduction to Software Project Management*

- Definition and overview
- Importance in software development

#### *2. Project Management Concepts*

- Core principles
- Roles and responsibilities in project management

#### *3. Project Management Tools*

- Tools used for planning, tracking, and reporting

#### *4. PMI's Knowledge Areas*

- Key areas defined by the Project Management Institute

#### *5. PMI Framework and Process Groups*

- Framework overview
- Process groups:
  - Initiating
  - Planning
  - Executing
  - Monitoring and Controlling
  - Closing

#### *6. Understanding Organizations*

- Structure and types of organizations
- Organizational influence on projects

#### *7. Project Planning*

- Goals and objectives
- Developing a project plan

#### *8. Project Evaluation*

- Techniques and criteria for evaluation

#### *9. Selection of an Appropriate Approach in Project*

- Traditional, agile, and hybrid approaches
- Criteria for selection

## *10. Software Effort Estimation*

- Methods of estimation:
  - Function points
  - Lines of code
  - COCOMO model

## *11. Activity Planning*

- Work breakdown structure (WBS)
- Scheduling techniques

## *12. Risk Management*

- Identifying risks
- Analyzing and mitigating risks

## *13. Evaluating the Risks to the Schedule*

- Impact of risks on timelines
- Tools for risk schedule analysis

## *14. Risk Control*

- Strategies for controlling risks
- Contingency planning

## *15. Configuration Management and Maintenance*

- Version control
- Maintenance processes

## *16. Environment for Configuration Control*

- Tools and best practices for configuration control

## *17. Resource Allocation*

- Techniques for allocating resources effectively
- Managing conflicts

## *18. Monitoring and Control*

- Performance measurement
- Change control processes

## *19. Review and Evaluation*

- Project reviews
- Post-mortem evaluation

## *20. Challenges of Outsourcing in Project Management*

- Risks and benefits of outsourcing
- Managing outsourced teams

